TEAM #45

PREDICTING CLAIMS & LOSSES FOR SMARTER AUTO INSURANCE PRICING

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1. Executive Summary

In this project, we developed predictive models to help insurers better manage pricing, underwriting, and customer retention by estimating individual-level risk. Our approach focused on two key outcomes: expected loss cost (LC and HALC) and claim likelihood (Claim Status).

For loss prediction, we tested multiple models and selected LightGBM based on its strong performance, achieving an R^2 of 0.7744 and a mean absolute error (MAE) of 14.14 for HALC. For claim classification, Gradient Boosting delivered the best results, achieving a ROC-AUC of 0.7889 on the test set.

We addressed class imbalance through weighting strategies and used SHAP-based feature interpretation and customer segmentation to connect modeling outputs to business strategy. Our recommendations include adjusting pricing for high-risk segments, prioritizing underwriting review, and implementing retention strategies targeted at low-risk, high-value customers.

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2. Introduction

In the insurance industry, accurately predicting the financial risk associated with new policyholders is crucial for premium pricing and overall risk management.

Task 1 focuses on predicting two cost-related metrics for new policyholders:

- **Loss Cost (LC)**: Incurred loss divided by exposure units.
- **Historically Adjusted Loss Cost (HALC)**: LC adjusted by a historical factor.

Task 2 is a binary classification problem: predicting whether a new policyholder will make a claim (**Claim Status, CS**), evaluated by **ROC-AUC**.

The goal is to build models that generalize well to unseen data, evaluated using **out-of-sample MSE/MAE** for regression targets and **ROC-AUC** for classification. We detail the steps taken for feature engineering, model selection, cross-validation, performance comparison, and final prediction.

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3. Data Preparation and Feature Engineering

We started with the provided training and test datasets, each comprising a wide array of date, categorical, and numeric fields. Several feature engineering steps were conducted:

- **Derived temporal features** such as age, license years, and contract start weekday/month/year.
- **Dropped irrelevant identifiers** (e.g., personal ID columns).
- **Label encoding** for categorical variables.
- **Median imputation** for missing values.

Two engineered targets were constructed using the following formulas:

- **LC = X.15 / X.16**
- **HALC = (X.15 / X.16) * X.18**

Due to their skewed distribution, both targets were log-transformed before modeling.

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4. Task 1: Loss Cost Prediction

To address the challenge of predicting highly skewed insurance losses, we experimented with four modeling approaches. The MLP model was trained using all available features without selection. For the GLM (Tweedie) model, we selected the top 8 features based on permutation importance. XGBoost and LightGBM models were each constructed using the top-ranked features derived from their respective feature importance metrics, and both were evaluated using 5-fold cross-validation to ensure model robustness.

Prior to modeling, we applied a log1p transformation to the target variables to reduce skewness and stabilize variance. Model training and evaluation were performed on the log scale, and predictions were subsequently converted back to the original scale via exponential transformation. This preprocessing step improved model stability and interpretability across all four approaches.

Based on the overall evaluation metrics, we selected the LightGBM model—one of the best-performing models—as the basis for generating the final predictions.

Model/ Metric (Real Scale)	Target	RMSE	MAE	MAPE	R2
MLP	LC	1263.1055	457.3748	151.09	-0.1078
	HALC	809.0291	259.7756	inf	0.9145
GLM	LC	1092.3193	423.466	130.06	0.1715
	HALC	4942.6269	1014.2173	inf	-0.0246
XGBoost	LC	1178.2798	414.9716	132.78	0.0563
	HALC	2087.8832	539.9814	112.63	0.0545
LightGBM	LC	1194.6484	421.4571	134.91	0.0521
	HALC	2072.0074	535.0635	111.92	0.0569

Figure 4.1: Performance of 4 models

Metric (Real Scale)	LC (Train)	LC (Predicted)	HALC (Train)	HALC (Predicted)
Mean	629.84	233.25	1156.17	325.79

Std	2448.34	78.55	4883.63	144.85

Figure4.2: Predictions(LightGBM)

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5. Task 2: Claim Classification

We explored seven models: Boosting and Boosting (weighted), Random Forest, Logistic Regression (Lasso/Ridge), and MLP. All models were evaluated with 5-fold CV using ROC-AUC as the key metric.

Model	Test ROC-AUC
Boosting	0.7889
Boosting (Weighted)	0.7888
Random Forest	0.7698
MLP	0.7659
Logistic Regression	0.7505
Logistic Regression (Lasso)	0.7457
Logistic Regression (Ridge)	0.7449

Figure 5.1: Test ROC-AUCs of 7 models

Final predictions from the best-performing boosting model yielded the following **confusion matrix:**

	Predicted 0	Predicted 1
Actual 0	6600	61
Actual 1	682	148

Figure 5.2: Confusion Matrix of Boosting (weighted) Model

Classification Report:

Class	Precision	Recall	F1-Score	Support
0	0.91	0.99	0.95	6661
1	0.71	0.18	0.28	830
Accuracy			0.90	7491
Macro Avg	0.81	0.58	0.62	7491
Weighted Avg	0.88	0.90	0.87	7491

Figure 5.3: Classification Report of Boosting (weighted) Model

Due to severe class imbalance in the data, we find that our model with the highest ROC-AUC only correctly classifies a small portion of the actual claims. At the small cost of about 0.001 in ROC-AUC, our weighted model still struggles, but has higher or equivalent values in all parts of the classification report. We most notably see improved performance in recall, where our normal boosting model only produced 0.06 for class 1. Thus, we choose this model, even though it does not have the highest test ROC-AUC.

6. Executive Summary

To support business understanding and enhance interpretability, we applied SHAP to our final LightGBM models. SHAP analysis identified X.8 (Insurance Years) and X.14 (Net Premium) as the most impactful predictors of LC. Higher premiums correlated with higher predicted losses, while Insurance Years showed a segmented effect, indicating risk differentiation by policyholder tenure.

6.1 Segment-Based Strategy (Insurance Years)

Insurance Segment	Mean LC	Std Dev (LC)	Count
New (0–2)	598.24	1653.74	1833
Emerging (3–5)	712.61	3434.75	1374
Established (6–10)	663.51	2412.66	601
Loyal (11+)	386.10	593.52	277

- **Emerging Segment (3–5 Years)**

This high-risk, volatile group shows the greatest average Loss Cost. We recommend:
➔ Risk Control Programs: Mobile-based education with incentive rewards.
➔ Usage-Based Insurance (UBI): Telematics-driven premium adjustments.
➔ Deductible Enforcement: Higher deductibles for those not meeting risk standards.

- **New Segment (0–2 Years)**

With large volume and moderate risk, this group benefits from:
➔ Selective Onboarding: Filter based on claim history and financial behavior.
➔ Progressive Coverage: Entry-level coverage with reward-based upgrades.
➔ Dynamic Pricing: Premiums adjusted based on initial behavioral data.

- **Loyal Segment (11+ Years)**

Low risk but underrepresented, deserving retention focus:
➔ Renewal Discounts: Incentivize long-term loyalty.
➔ Referral Incentives: Encourage customer-driven acquisition.
➔ Preventive Benefits: Provide added value via check-ups and safety perks.

6.2 Net Premium(X.14) segment analysis and strategic recommendations

Premium Segment	Mean LC	Std Dev (LC)	Count
Low (0–300)	519.80	2993.37	1753
Medium (300–800)	699.23	1977.00	2314
High (800–2000)	1014.69	1058.41	84

Very High (2000+)	—	—	0

- **Low Premium (0–300)**

Risk Adjustment Needed / High variability in Loss Cost (σ = 2993), possible underpricing.
➔ Tiered Deductibles: Scale with claim history.
➔ Behavior-Based Pricing: Reflects usage and habits.
➔ Selective Underwriting: Filter by claim-free or good record.

- **High Premium (800–2000)**

Retention & Upsell Opportunity / High average LC (1014.7), but stable risk (σ = 1058).
➔ Predictable Plans: Stable premiums + enhanced coverage.
➔ Value-Added Benefits: Concierge, quick claims, roadside help.
➔ Proactive Monitoring: Dedicated review service.

- **Very High Premium (2000+)**

Untapped Market / No current customers. Indicates expansion potential.
➔ Premium Product Launch: High-limit, VIP-tier coverage.
➔ Targeted Campaigns: Aim at high-net-worth profiles.
➔ Performance-Based Rebates: Loyalty/low-claim incentives.

7.Future Work

To maximize business impact, future work should focus on operationalizing model outputs:

- Dynamic Pricing: Integrate predicted LC and HALC into real-time premium adjustments.
- Risk Flagging: Automate manual review triggers for high-risk customers based on CS and HALC.
- Retention Strategy: Target loyalty programs to low-risk segments identified through modeling.
- Model Maintenance: Periodically retrain models with updated behavioral and claims data.

These steps will align pricing, underwriting, and retention more closely with customer-level risk, improving portfolio profitability.

8.Conclusion

In this project, we built predictive models to estimate individual insurance risk by forecasting both claim probability and expected loss cost. For Task 1 (Loss Cost Prediction), we evaluated four models—MLP, GLM (Tweedie), XGBoost, and LightGBM—each using tailored feature selection strategies. MLP used all features, GLM used the top 8 features from permutation importance, and the tree-based models used top-ranked features from internal feature importances. Due to heavy skewness in the targets (LC and HALC), we applied log1p transformation prior to training and reverted predictions to the original scale with expm1, stabilizing variance and improving interpretability. Among these, LightGBM achieved the best overall performance in both LC and HALC, and was selected for generating final predictions.

For Task 2 (Claim Classification), we tested several models including Boosting, Random Forest, and Logistic Regression. The best-performing model—Boosting with weighting—achieved an ROC-AUC of 0.7888, showing strong ability to distinguish claim vs. no-claim policyholders.

To support business understanding, we applied SHAP analysis to interpret model behavior and identify key drivers of loss. Variables like Net Premium (X.14) and Insurance Years (X.8) emerged as the most impactful predictors. These insights enabled segment-based recommendations, such as rewarding loyal customers, targeting new or emerging segments with tiered pricing, and designing usage-based insurance plans.

Our findings provide actionable strategies for risk-adjusted pricing, retention incentives, and dynamic underwriting in real-world insurance operations.

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9. Appendix

9.1 Task 1

Appendix A: Model Comparison and Evaluation
This section summarizes the performance of all tested models—GLM, MLP, XGBoost, and LightGBM—using 5-fold cross-validation. Each model used a distinct feature selection approach: MLP used all available features, GLM was trained on the top 8 features identified via permutation importance, while XGBoost and LightGBM used their own internal feature importance rankings. To address the skewness of the LC and HALC targets, a log1p transformation was applied during training, and predictions were converted back to the original (real) scale using expm1 for evaluation. Performance was assessed based on R^2, Mean Squared Error (MSE), and Mean Absolute Error (MAE) in the original scale.

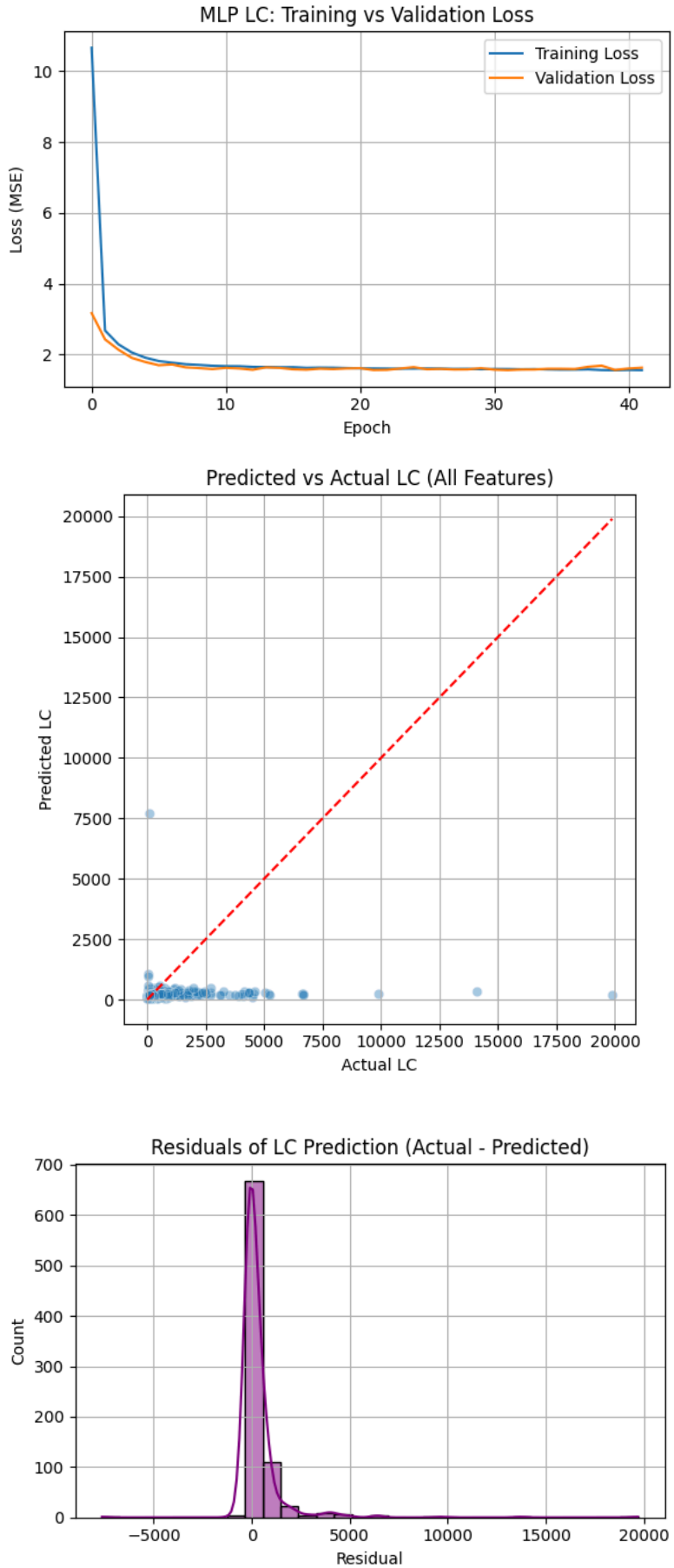


Figure A1: MLP LC Prediction

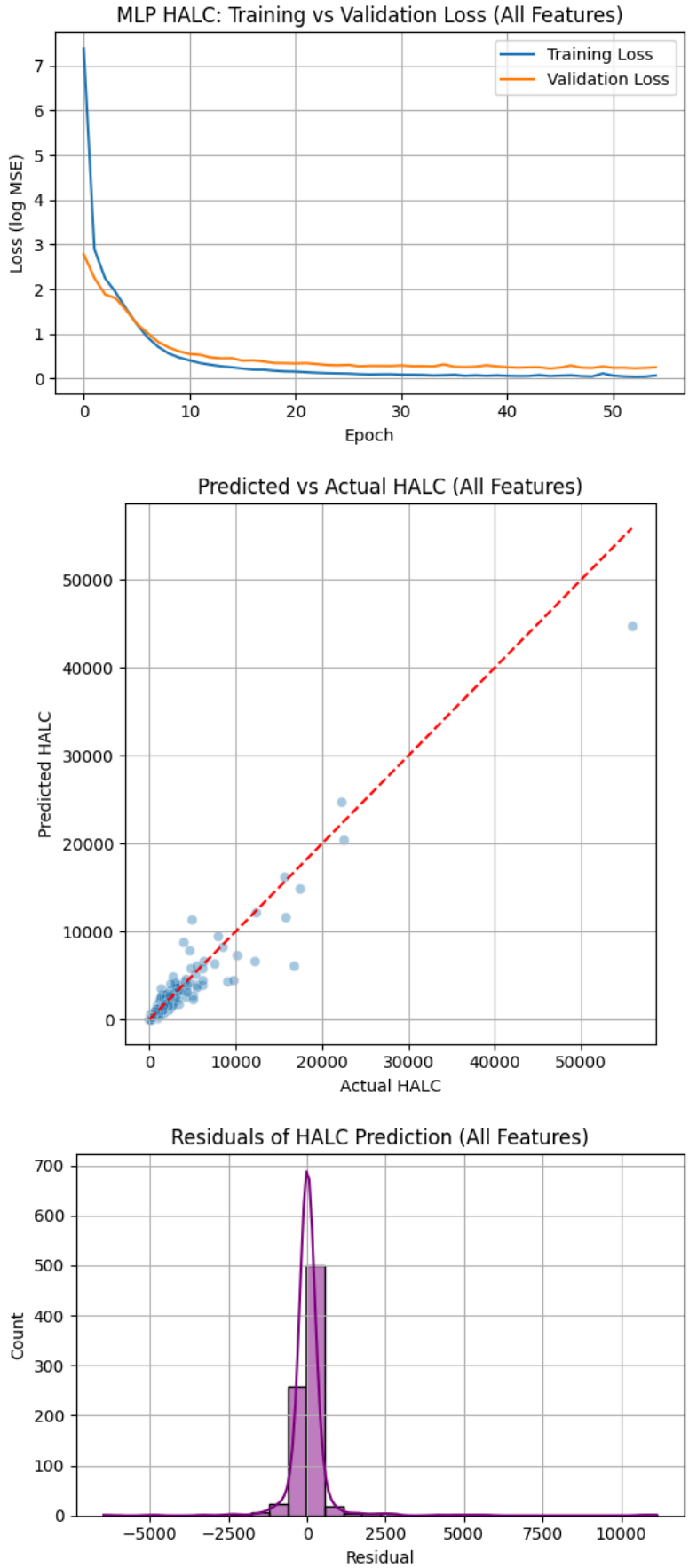


Figure A2: MLP HALC Prediction

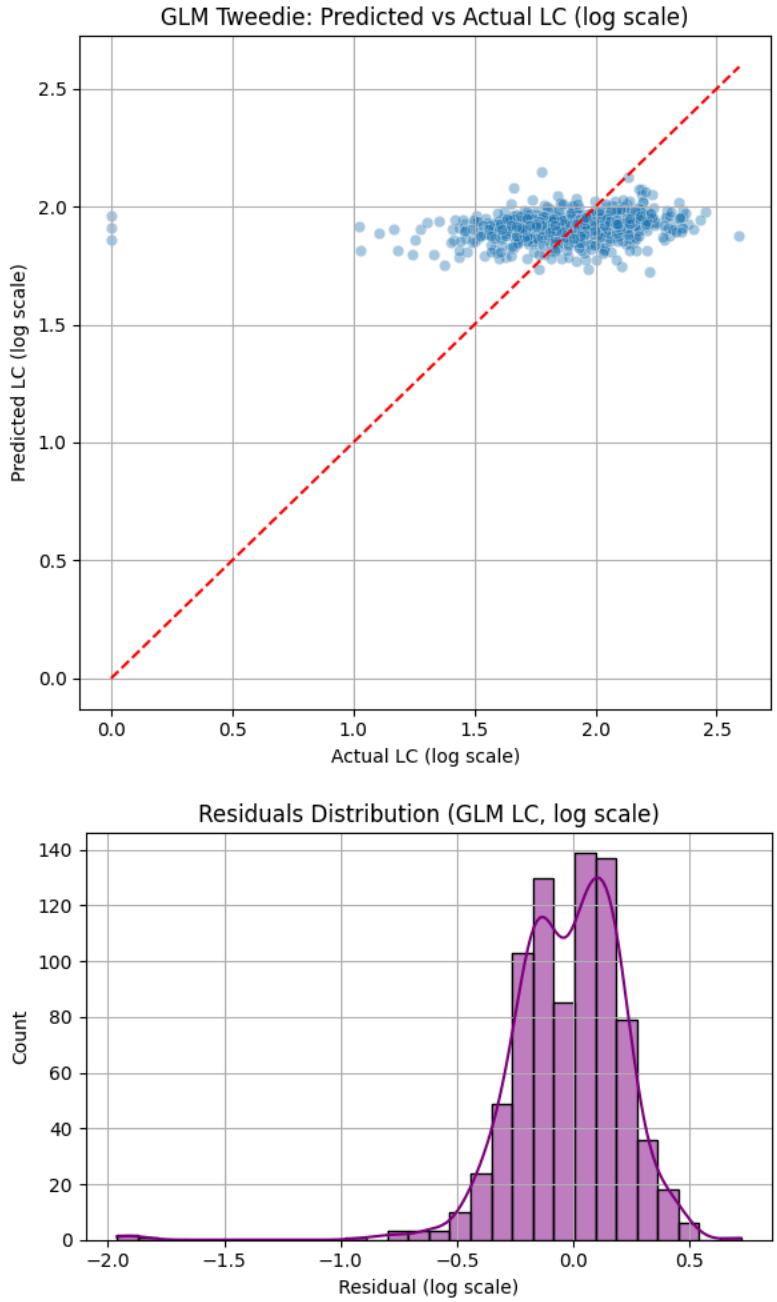


Figure A3: GLM: Tweedie LC Prediction

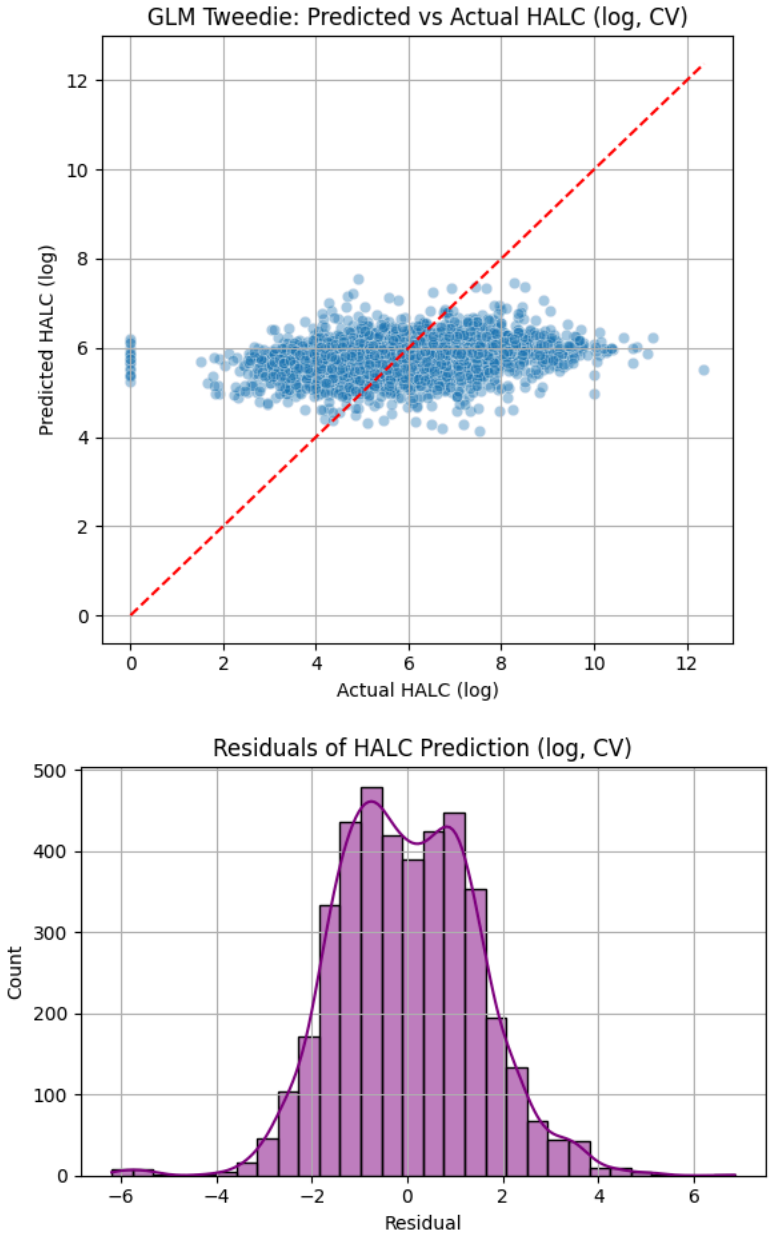
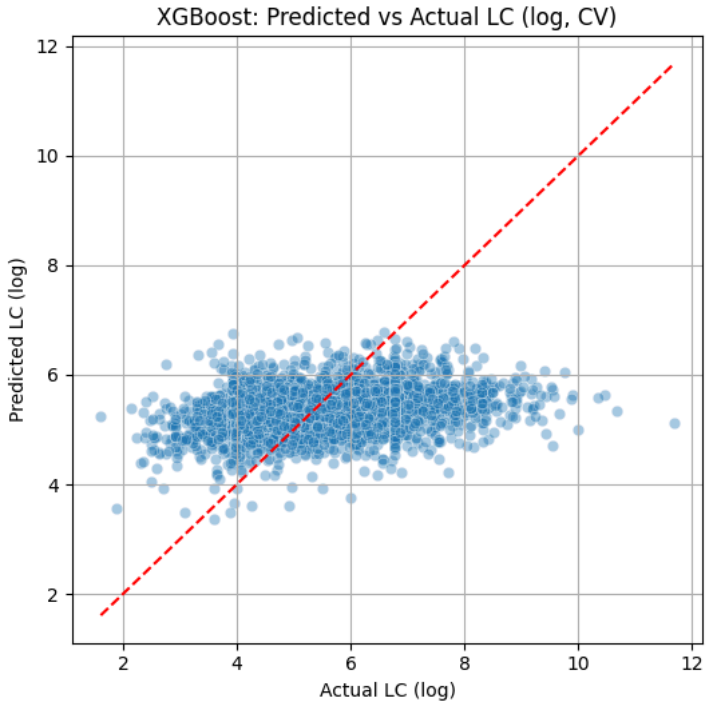


Figure A4: GLM: Tweedie HALC Prediction



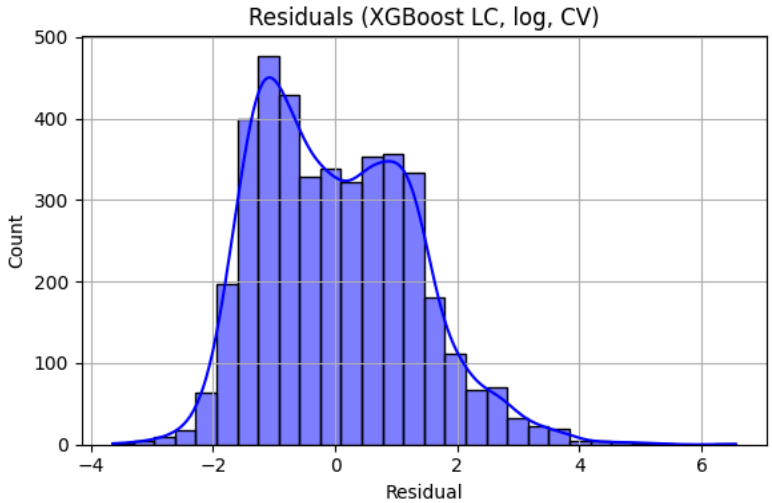


Figure A5: XGBoost (objective='tweedie'LC Prediction

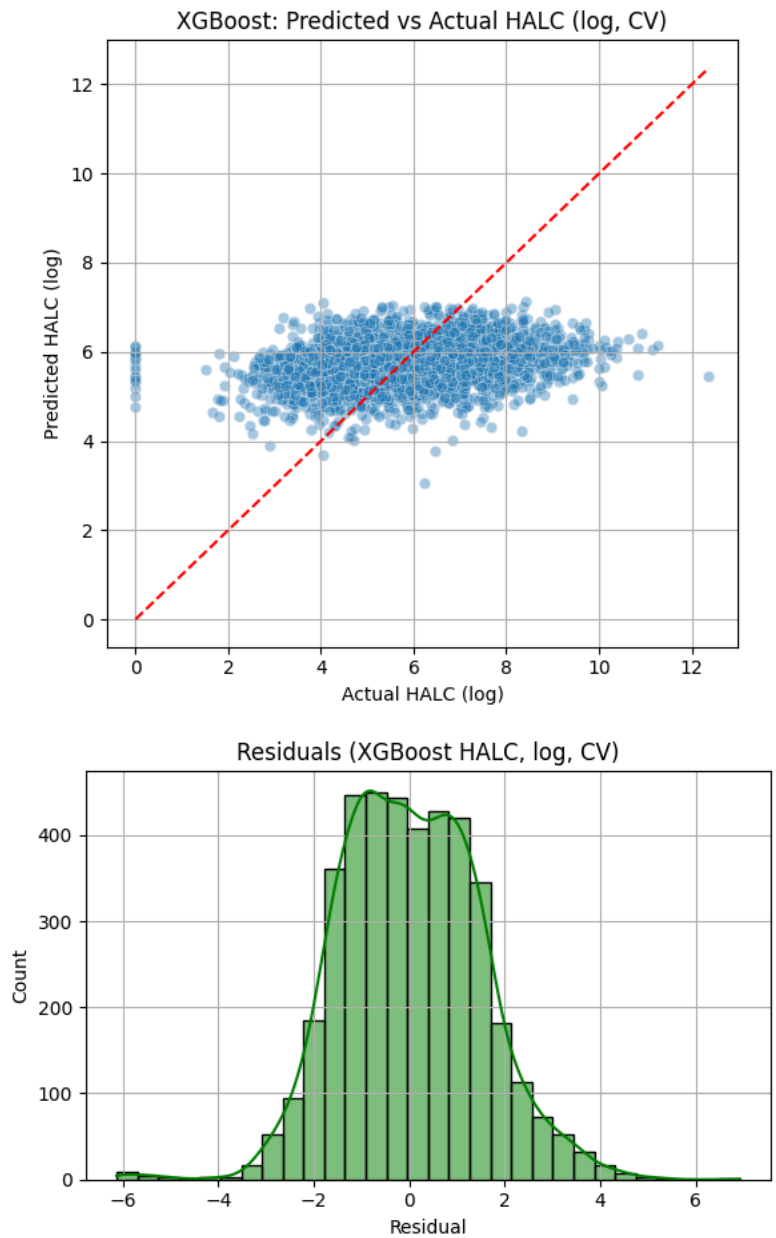


Figure A6: XGBoost (objective='tweedie') HALC Prediction

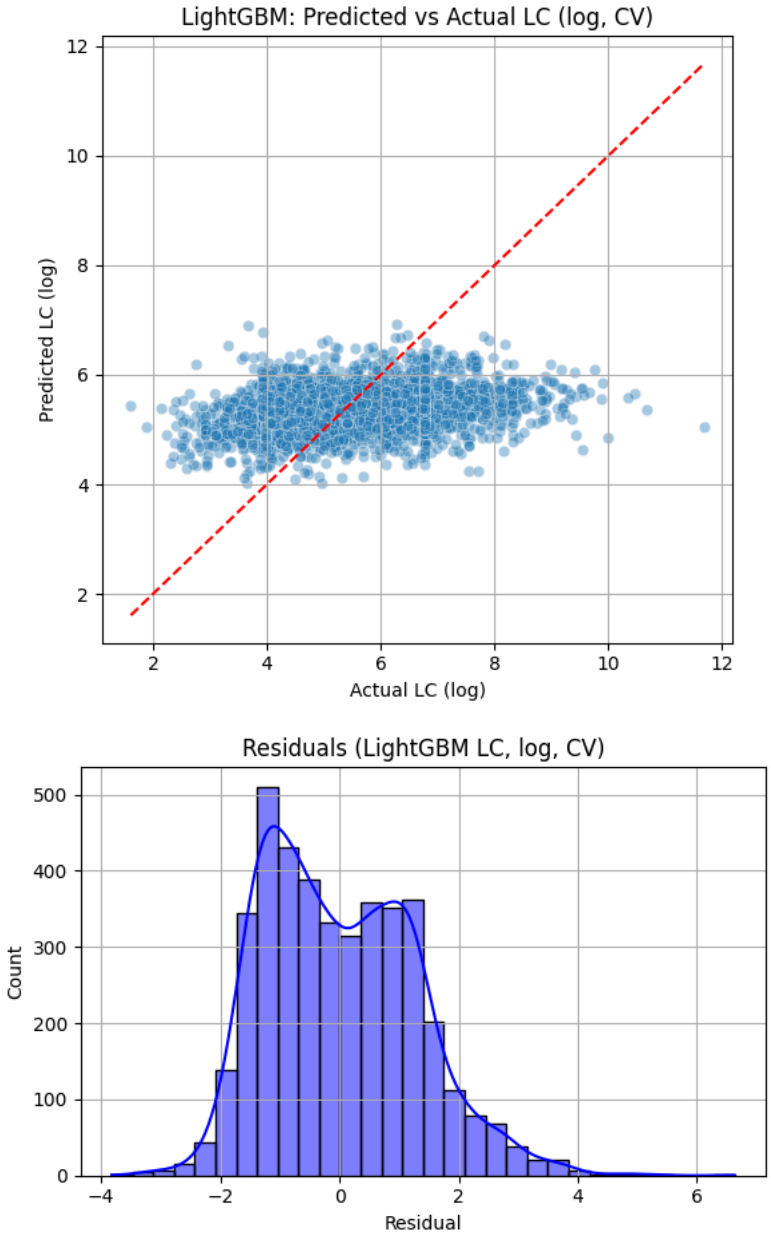


Figure A7: LightGBM (objective='tweedie') LC Prediction

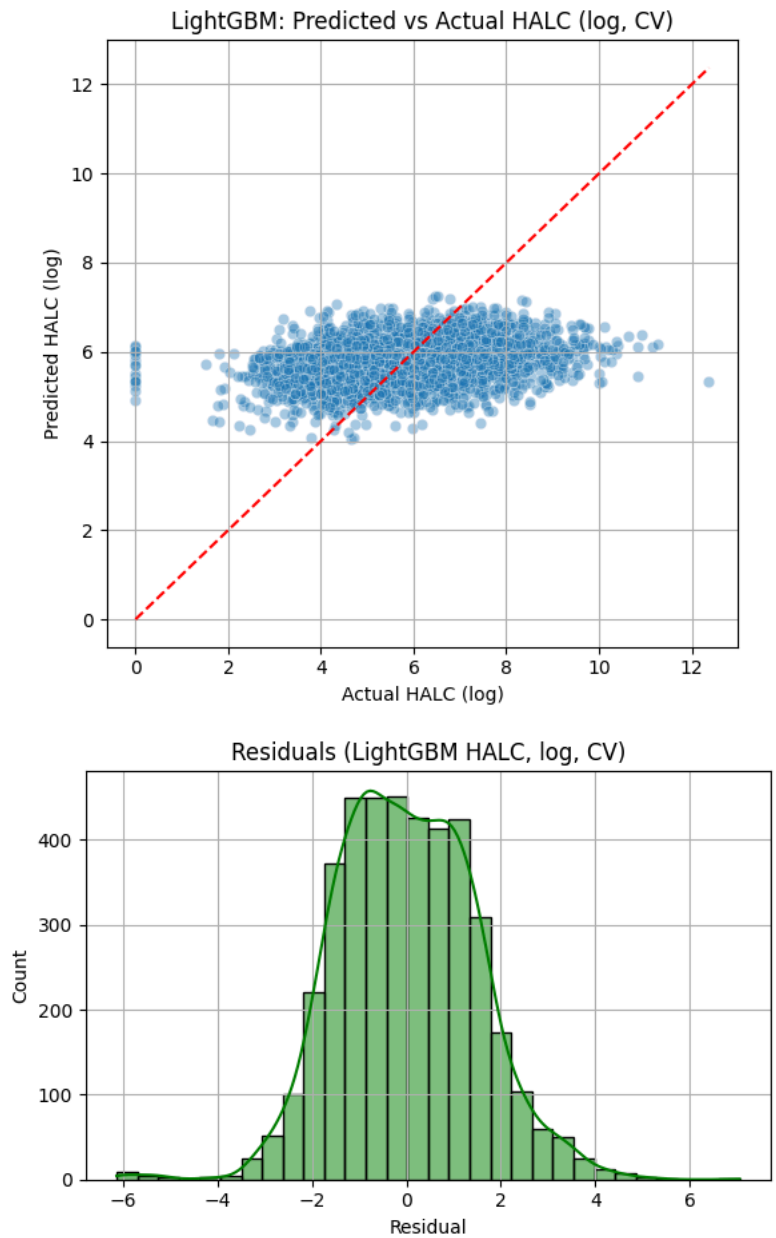


Figure A8: LightGBM (objective='tweedie') HALC Prediction

Appendix B: Feature Importance Analysis

Here, we provide the top features selected based on XGBoost-derived importance scores. These were used consistently across all models except for GLM, which relied on permutation importance for feature selection. Notably, XGBoost and LightGBM produced similar feature importance rankings, supporting the stability and consistency of key predictors across tree-based models. Feature importance bar plots are included to highlight the most predictive variables..

✅ Permutation Feature Importance (GLM - LC log target):

	Feature	Importance	Std
8	X.15	0.144303	0.007992
24	policy_duration	0.105087	0.010653
28	contract_year	0.085702	0.006923
9	X.16	0.036663	0.008433
2	X.9	0.034867	0.006491
7	X.14	0.024088	0.006594
25	vehicle_age	0.021247	0.003656
5	X.12	0.017484	0.002994
18	X.25	0.006193	0.003247
11	X.18	0.004767	0.002210
10	X.17	0.003647	0.000987
13	X.20	0.001453	0.001407
26	contract_month	0.001053	0.001038
1	X.8	0.001025	0.000506
14	X.21	0.000783	0.001190
23	license_years	0.000741	0.000437
6	X.13	0.000724	0.001089
17	X.24	0.000165	0.000748
21	X.28	-0.000080	0.000047
15	X.22	-0.000128	0.000060
22	age	-0.000189	0.000122
0	X.7	-0.000359	0.000834
27	contract_weekday	-0.000417	0.000572
20	X.27	-0.000684	0.001075
19	X.26	-0.000733	0.000536
3	X.10	-0.001040	0.002512
4	X.11	-0.001432	0.000567
12	X.19	-0.001722	0.000816
16	X.23	-0.003371	0.003334

Figure B1: GLM: Permutation

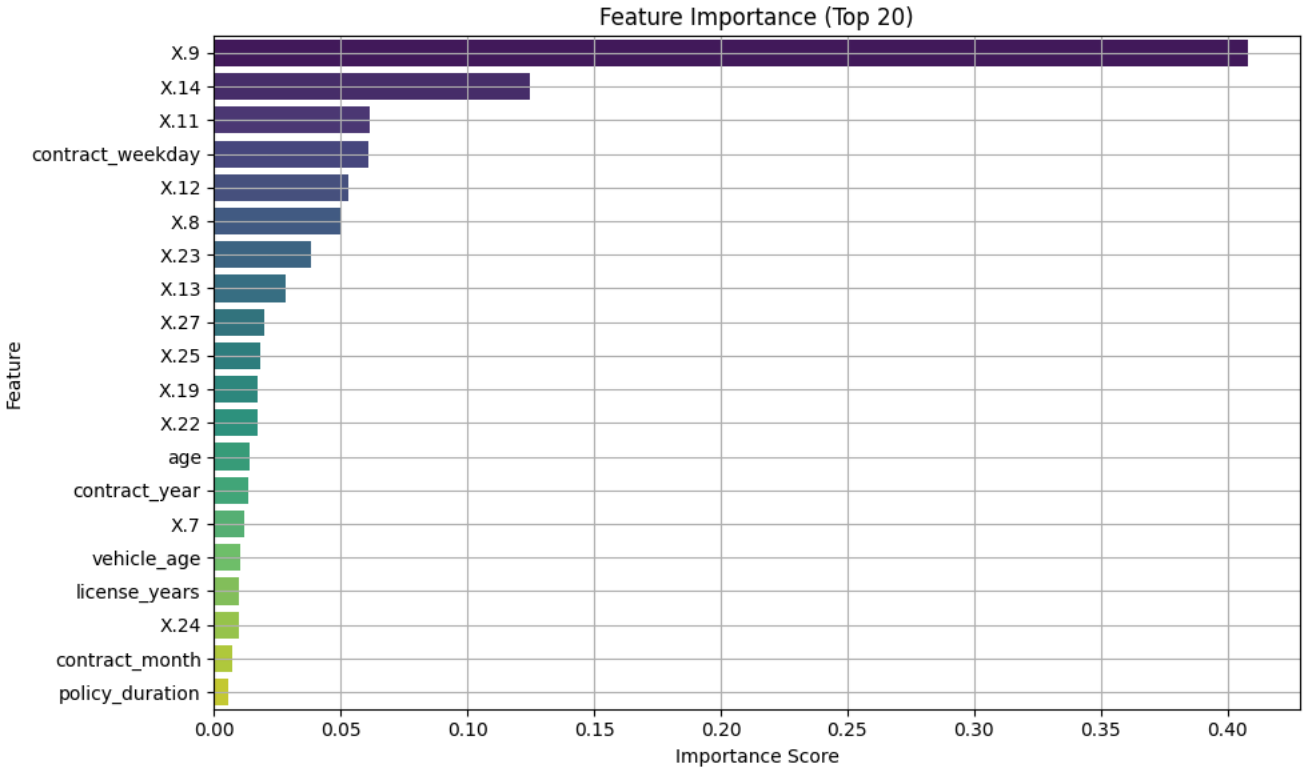


Figure B2: XGBoost Feature Importance

Appendix C: SHAP-Based Model Interpretation

This section presents the SHAP (SHapley Additive exPlanations) analysis conducted on the final LightGBM model. Both the mean absolute SHAP values and summary plots are included to visualize feature influence and directionality.

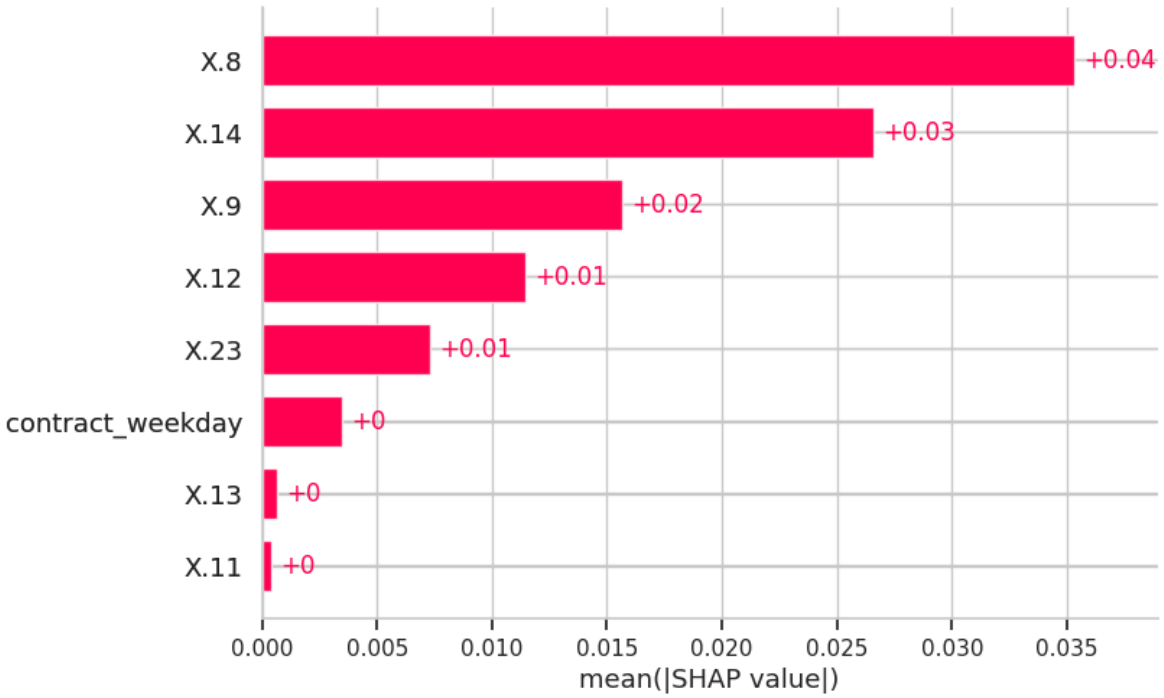


Figure C1: Mean SHAP Value of LightGBM

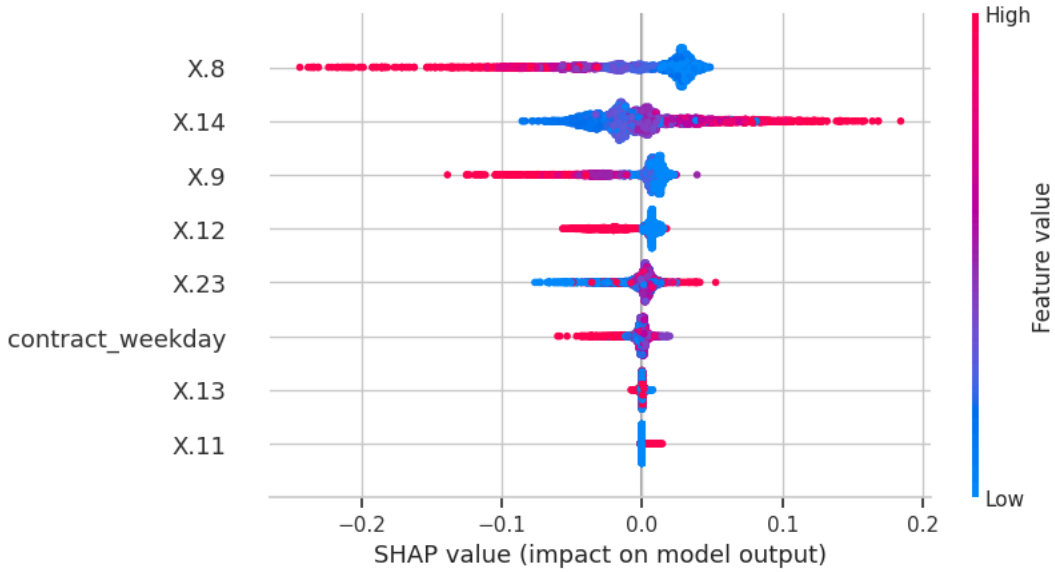
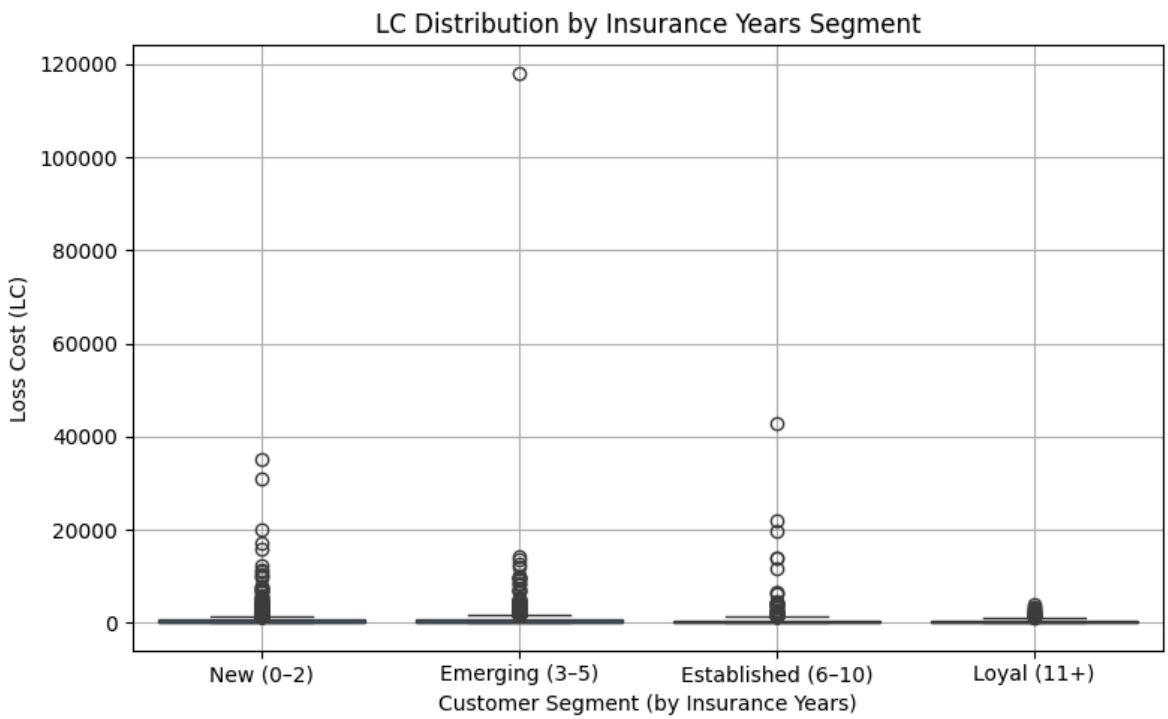


Figure C2: Feature Value of LightGBM

Appendix D: Segmentation Analysis

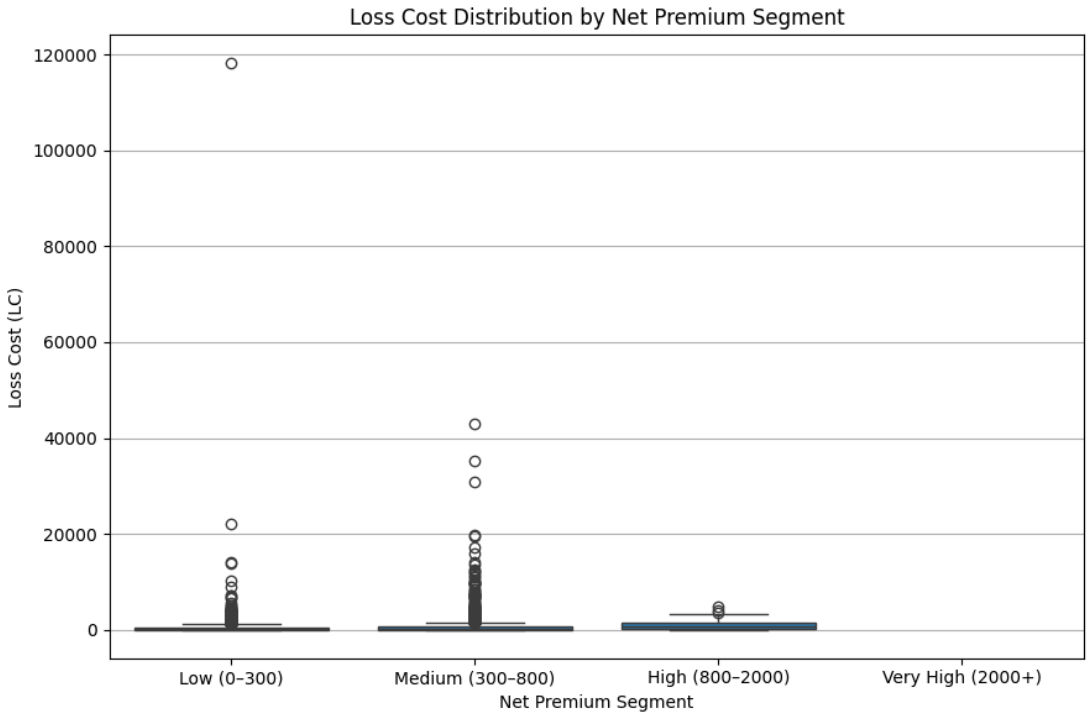
This section illustrates customer segmentation based on two key features:

Insurance Tenure (X.8): Segments include New, Emerging, Established, and Loyal groups, with boxplots and summary stats showing differing loss profiles. Net Premium (X.14): Premium tiers are analyzed in relation to loss cost to reveal underpricing risk and retention opportunities.



Insurance Segment	Mean LC	Std Dev LC	Count
New (0–2)	598.24	1653.74	1833
Emerging (3–5)	712.61	3434.75	1374
Established (6–10)	663.51	2412.66	601

Loyal (11+)	386.1	593.52	277



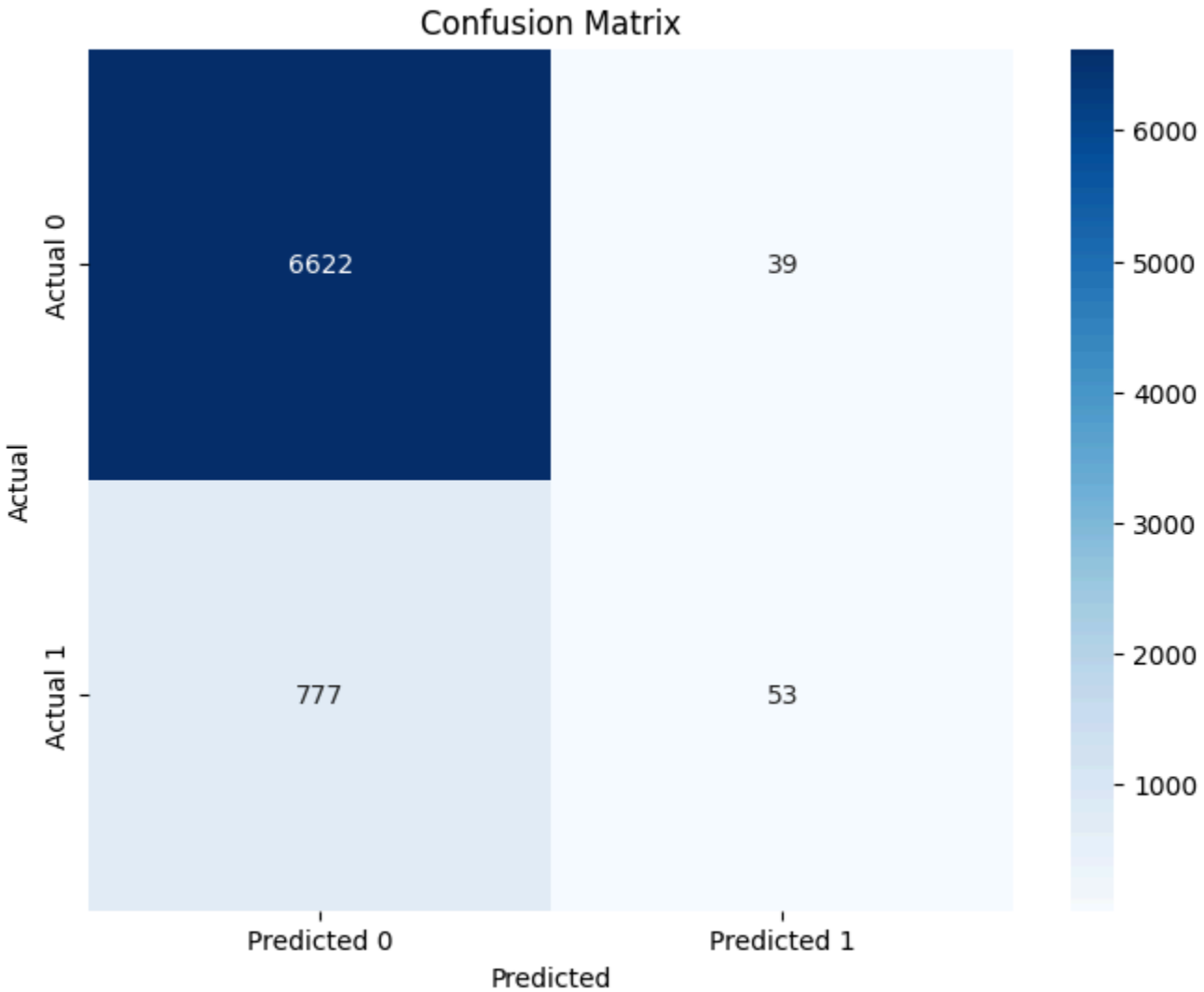
premium_segment	mean	std	count
Low (0–300)	519.8	2993.37	1753
Medium (300–800)	699.23	1977	2314
High (800–2000)	1014.69	1058.41	84
Very High (2000+)			0

9. Appendix

9.2 Task 2

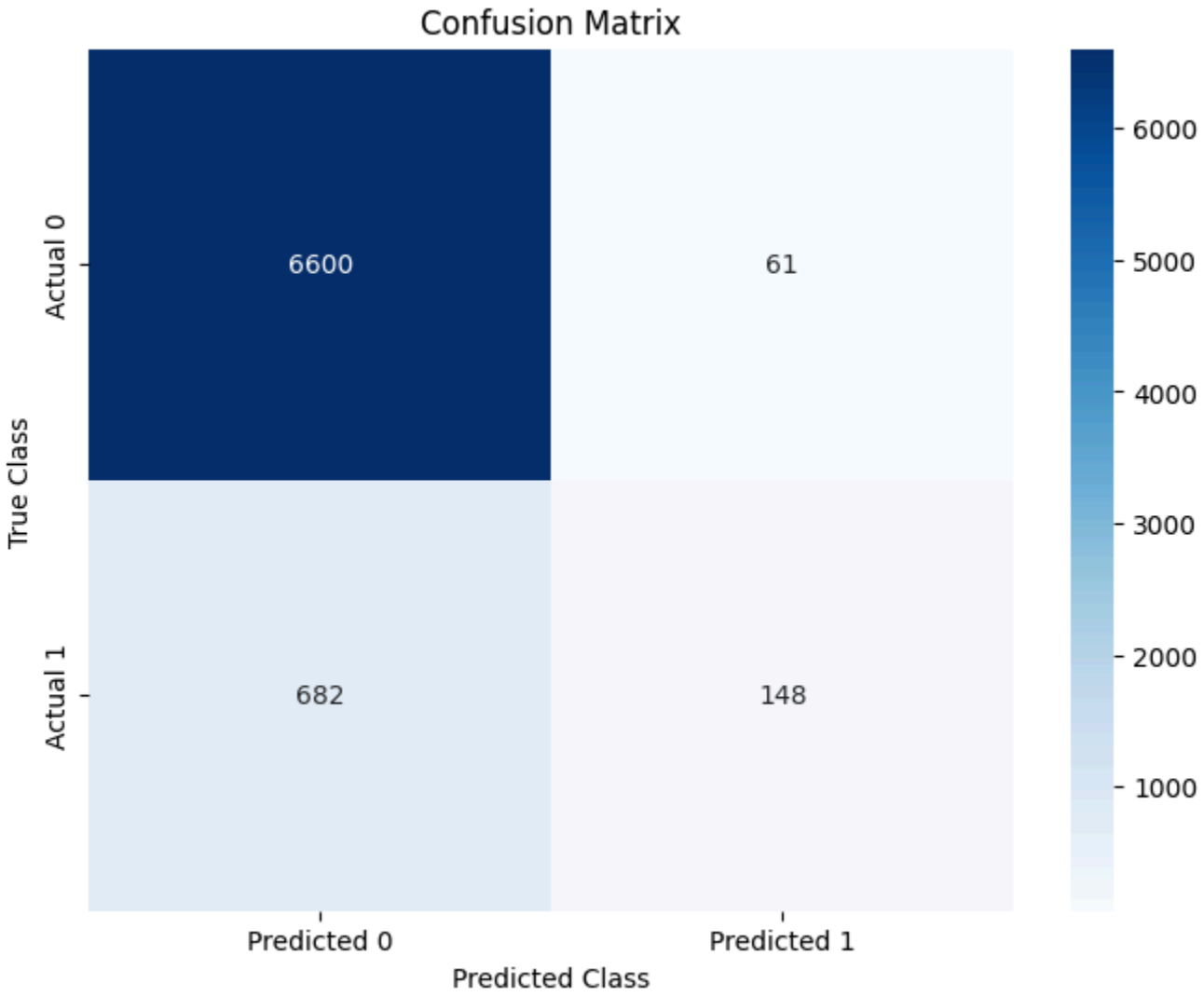
Appendix A: Model Comparison and Evaluation

We performed 5 fold cross validation when training each model. Each model was also trained using grid search in order to attempt to find optimal hyperparameters for each model. We attempted feature selection, but, other than in random forest, the models performed best with all features. Listed below is the performance of each of our trained models in the same order as in our ROC-AUC table before. Models include a confusion matrix and classification report since ROC-AUC is already reported earlier.



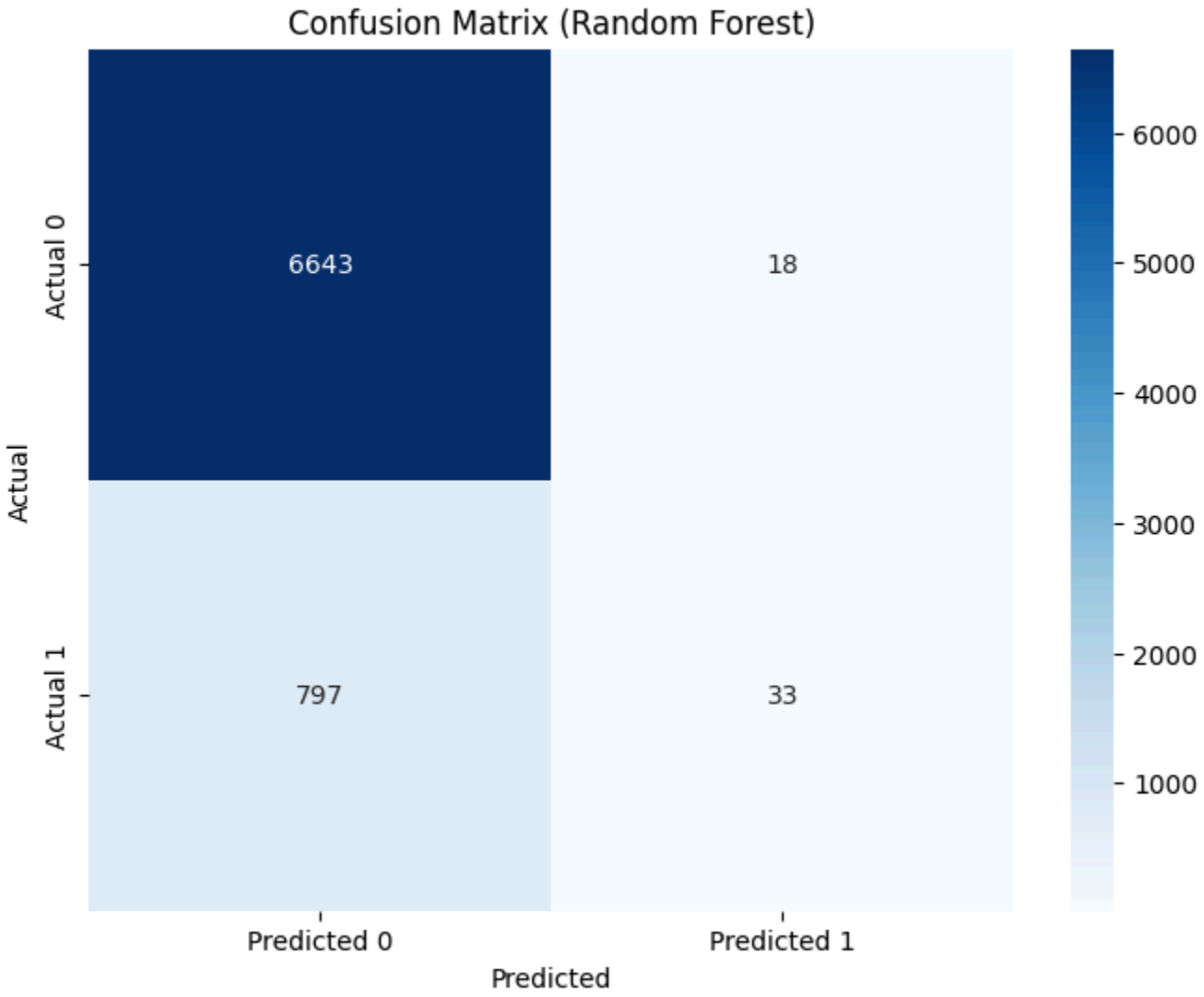
	precision	recall	f1-score	support
0	0.89	0.99	0.94	6661
1	0.58	0.06	0.11	830
accuracy			0.89	7491
macro avg	0.74	0.53	0.53	7491
weighted avg	0.86	0.89	0.85	7491

Figure A1: Boosting Confusion Matrix and Classification Report



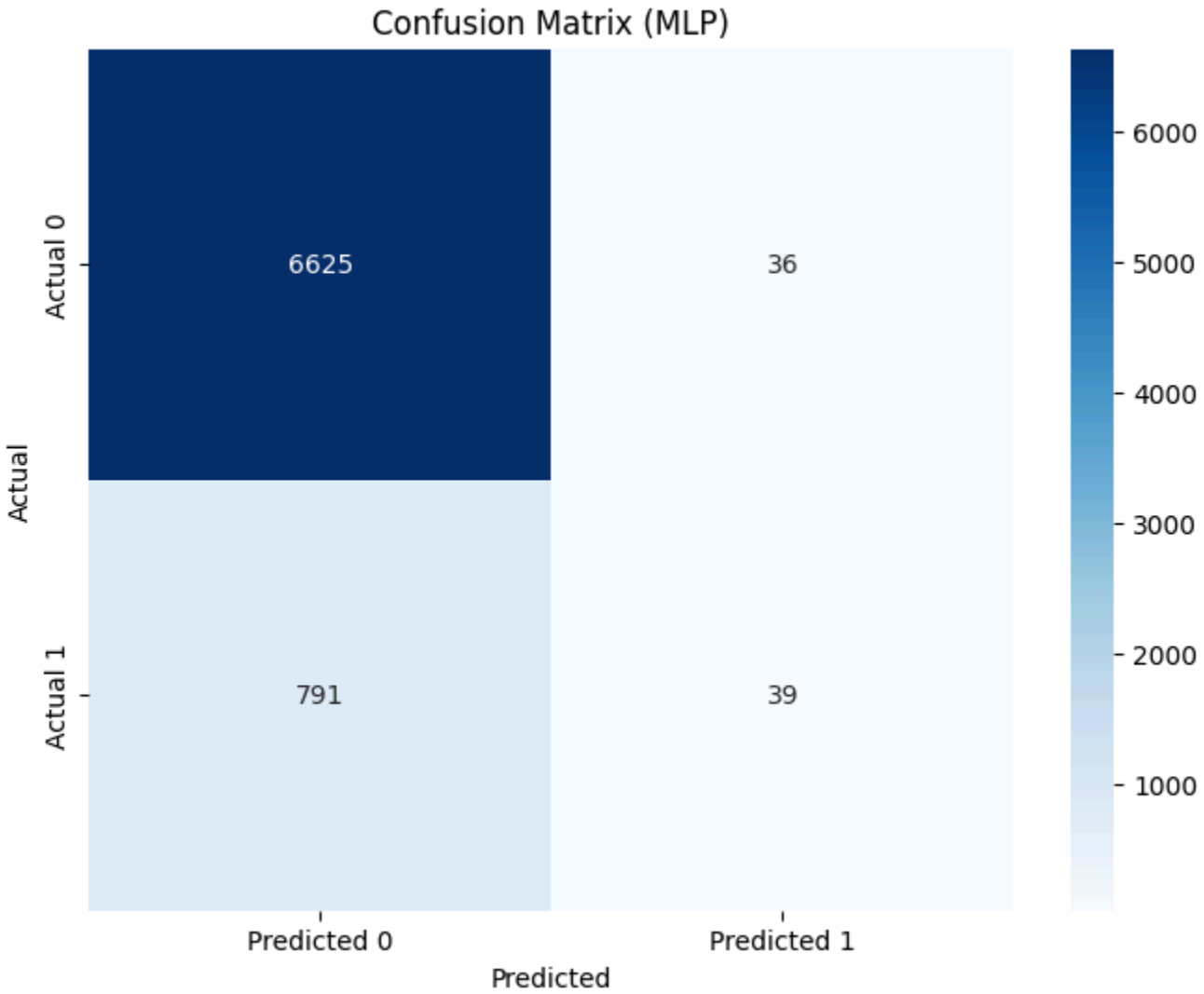
	precision	recall	f1-score	support
0	0.91	0.99	0.95	6661
1	0.71	0.18	0.28	830
accuracy			0.90	7491
macro avg	0.81	0.58	0.62	7491
weighted avg	0.88	0.90	0.87	7491

Figure A2: Boosting (Weighted) Confusion Matrix and Classification Report



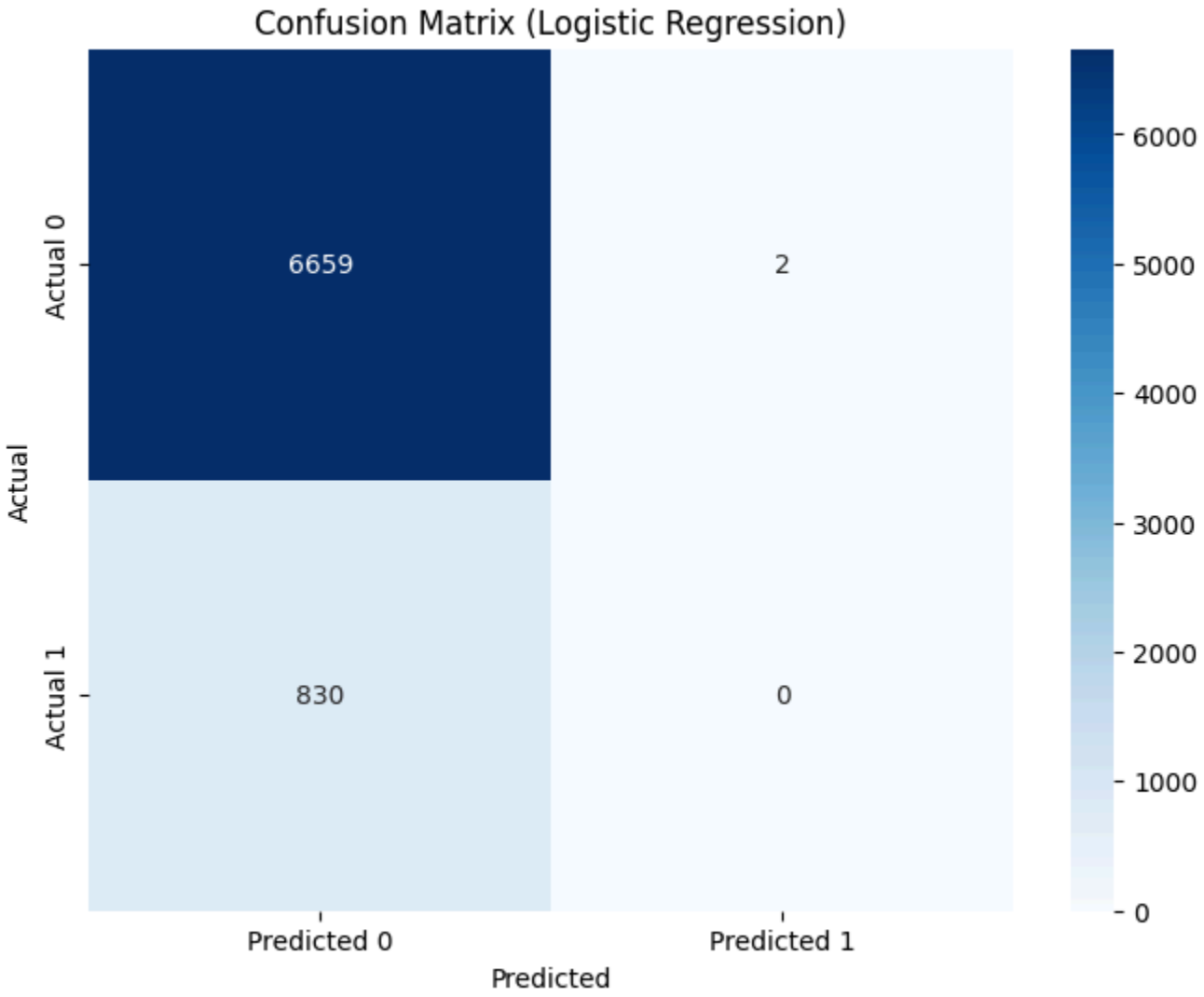
	precision	recall	f1-score	support
0	0.89	1.00	0.94	6661
1	0.65	0.04	0.07	830
accuracy			0.89	7491
macro avg	0.77	0.52	0.51	7491
weighted avg	0.87	0.89	0.85	7491

Figure A3: Random Forest Confusion Matrix and Classification Report



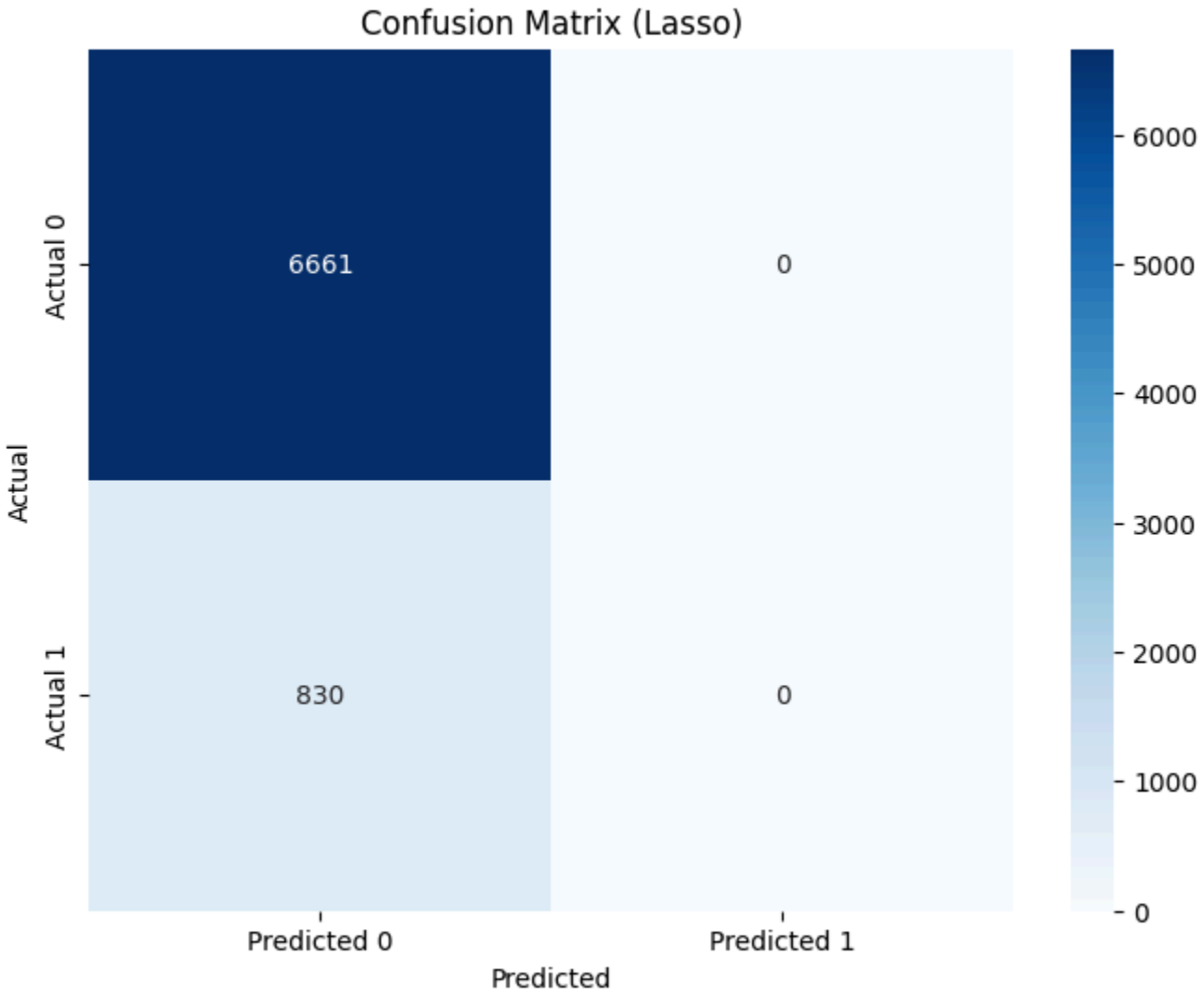
	precision	recall	f1-score	support
0	0.89	0.99	0.94	6661
1	0.52	0.05	0.09	830
accuracy			0.89	7491
macro avg	0.71	0.52	0.51	7491
weighted avg	0.85	0.89	0.85	7491

Figure A4: MLP Confusion Matrix and Classification Report



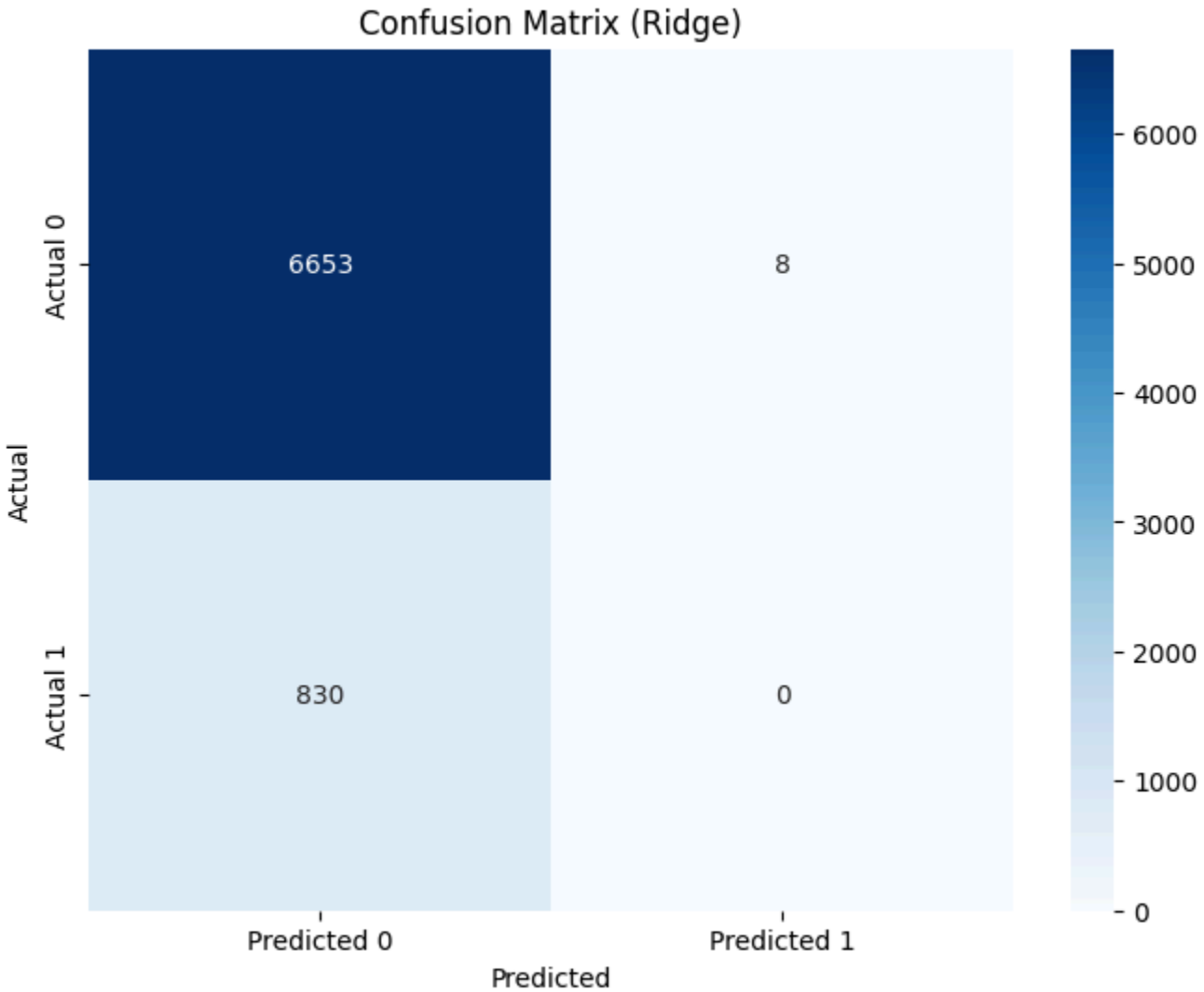
	precision	recall	f1-score	support
0	0.89	1.00	0.94	6661
1	0.00	0.00	0.00	830
accuracy			0.89	7491
macro avg	0.44	0.50	0.47	7491
weighted avg	0.79	0.89	0.84	7491

Figure A4: Logistic Regression Confusion Matrix and Classification Report



	precision	recall	f1-score	support
0	0.89	1.00	0.94	6661
1	0.00	0.00	0.00	830
accuracy			0.89	7491
macro avg	0.44	0.50	0.47	7491
weighted avg	0.79	0.89	0.84	7491

Figure A5: Lasso Confusion Matrix and Classification Report



	precision	recall	f1-score	support
0	0.89	1.00	0.94	6661
1	0.00	0.00	0.00	830
accuracy			0.89	7491
macro avg	0.44	0.50	0.47	7491
weighted avg	0.79	0.89	0.84	7491

Figure A6: Ridge Confusion Matrix and Classification Report

Appendix B: SHAP Values and Customer Segmentation

These plots helped to drive business interpretation by showing which features contributed the most to higher risk of filing a claim. In particular, segmentation and the dependence plot show that our newest customers tend to be the riskiest.

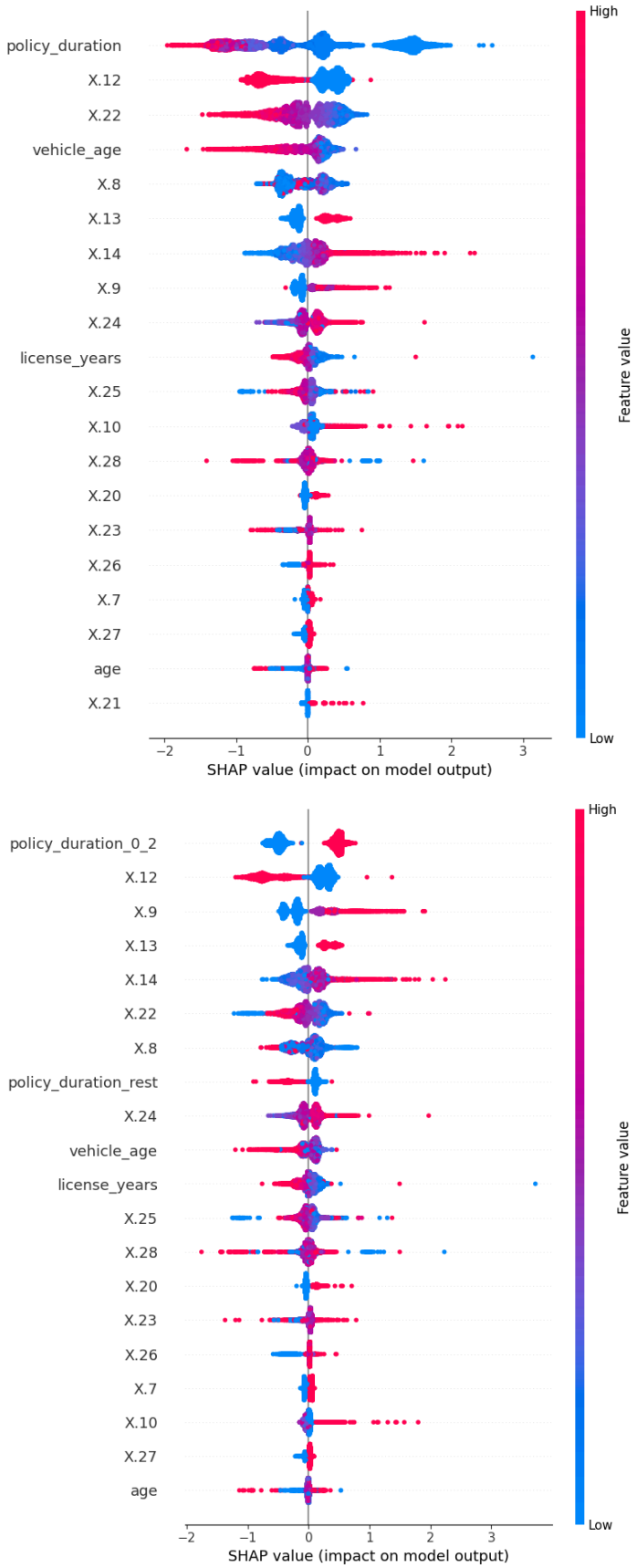


Figure B1: SHAP Value plots. Second plot has segmented policy duration.

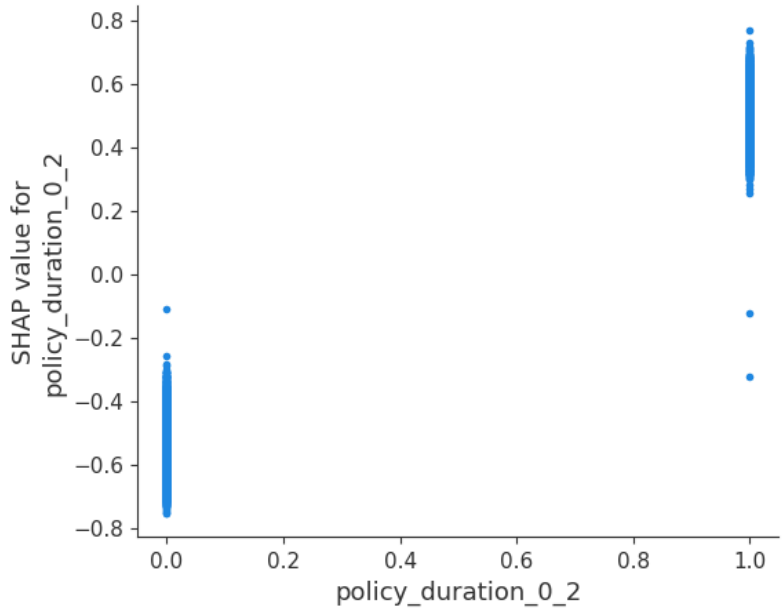


Figure B2: SHAP Dependence Plot of Policy Duration 0-2 Years